Wildland surface fire **spread** modelling, 1990–2007. 2: Empirical and quasi-empirical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… **spread** for the prevailing conditions (the **prediction** of which is the primary aim of all fire **spread prediction** … covered in the present **review** as it is outside the scope of surface fire **spread**). …

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Wildfire prediction to inform fire **management**: statistical science challenges

SW Taylor, DG Woolford, CB Dean, DL Martell - 2013 - projecteuclid.org

… In this paper we **review** some of the models that have been developed to predict fire occurrence, … **spread model** against **wildfire** observations. A major challenge in this regard is that …

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Evaluating a **model** for predicting active crown fire rate of **spread** using **wildfire** observations

ME Alexander, MG Cruz - Canadian Journal of Forest Research, 2006 - cdnsciencepub.com

… With 57 quantitative **wildfire** observations, no study to date … to evaluate a predictive fire **behavior model**. Thus, contrary to … — A **review** of fire weather, fuels, and fire **behavior**. Can. For. …

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Review of **wildfire management** techniques—Part I: Causes, prevention, detection, suppression, and data analytics

S Jazebi, F De Leon, A Nelson - IEEE Transactions on Power …, 2019 - ieeexplore.ieee.org

… in consecutive frames over time for remote monitoring of **wildfire spread**. In [45], a … **forest fire** with low consumption of energy. The fire **spread model** in [46] can produce fire **propagation** …

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On the need for a theory of **wildland fire spread**

MA Finney, JD Cohen, SS McAllister… - … journal of **wildland fire**, 2012 - CSIRO Publishing

… In summary, our **review** of existing research indicates an insufficient experimental basis for … an insufficient basis for related fire **spread model** assumptions. Independent experimental …

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Modeling **wildland fire propagation** with level set methods

V Mallet, DE Keyes, FE Fendell - Computers & Mathematics with …, 2009 - Elsevier

… burning, and post-crisis **review**, eg, personnel training, … **wildland fire spread** models, especially a semi-empirical, equilibrium-type **model** proposed in [11] for wind-aided fire **spread** …

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A **review** on status, implications and recent trends of **forest fire management**

V Attri, R Dhiman, S Sarvade - Archives of Agriculture and …, 2020 - academia.edu

… **Forest fire spread** out in an area having combustible material … **Forest fire** starts from the ignition point and **spread** outwards … topography were made favorable environment for **forest fire**. …

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A **review** of recent forest and **wildland fire management** decision support systems research

DL Martell - Current Forestry Reports, 2015 - Springer

… Fire behaviour or the physics and chemistry of fire ignition and **spread**. … focus of this **review** is very decidedly North America for the period 2010 through 2014. **Wildland fire** and the …

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Forest fire spread simulation and fire extinguishing visualization research
Q Meng, H Lu, Y Huai, H Xu, S Yang - Forests, 2023 - mdpi.com
… **forest fire spread** quickly. Therefore, this paper proposes a lightweight **forest fire spread model** … ensuring the realism of **forest fire behavior** and the immersion of the virtual environment. …
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Forest fire spread simulating **model** using cellular automaton with extreme learning machine
Z Zheng, W Huang, S Li, Y Zeng - Ecological Modelling, 2017 - Elsevier
… **simulation** of **forest fire** spreading plays an essential role in designing quick risk **management** and … By integrating the ELM with the traditional **forest fire** CA framework, a new cellular …
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WRF-Fire: coupled weather–**wildland fire** modeling with the weather research and forecasting **model**
JL Coen, M Cameron, J Michalakes… - Journal of Applied …, 2013 - journals.ametsoc.org
… numerical weather **prediction model**. The fire module is a surface fire-**behavior model** that is two-way coupled with the atmospheric **model**. Near-surface winds from the atmospheric …
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Wildland fire spread modeling using convolutional neural networks
JL Hodges, BY Lattimer - Fire technology, 2019 - Springer
… **spread** requires multiple simulations that may limit the number of scenarios which can be analyzed during an emergency. A new **wildland fire spread model** … **wildland fire spread model** …
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Traffic modeling for wildland–urban interface fire evacuation
P Intini, E Ronchi, S Gwynne, A Pel - Journal of Transportation …, 2019 - ascelibrary.org
… Based on this **review** analysis, recommendations are made … possible integrations with **wildfire** and pedestrian models. … and fire **propagation** speed may not affect the **model** choice. …
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A method for ensemble **wildland fire simulation**
MA Finney, IC Grenfell, CW McHugh, RC Seli… - … Modeling & Assessment, 2011 - Springer
… An ensemble **simulation** system that accounts for uncertainty in long-range weather conditions and two-dimensional **wildland fire spread** is described. Fuel moisture is expressed based …
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Review of **wildfire** modeling considering effects on land surfaces
D Or, E Furtak-Cole, M Berli, R Shillito… - Earth-Science …, 2023 - Elsevier
… **review** are: (1) to introduce hydrologists, soil scientists and ecologists to principles and advances in modeling of **wildfire dynamics** and rates of **spread** … large eddy **simulation** models. We …
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Review of vortices in **wildland fire**
JM Forthofer, SL Goodrick - Journal of Combustion, 2011 - Wiley Online Library
… fire **behavior** and safety concerns. In this paper, the current state of knowledge of the interaction of **wildland fire** … In this study a computer **model** of the atmospheric boundary layer was …
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A cellular automaton **model** of **wildfire propagation** and extinction
KC Clarke, JA Brass, PJ Riggan - … Engineering and Remote …, 1994 - research.fs.usda.gov
… We propose a new **model** to predict the spatial and temporal **behavior** of wildfires. Fire **spread** and intensity were simulated using a cellular automaton **model**. Monte Carlo techniques …
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Emulation of **wildland fire spread simulation** using deep learning

F Allaire, V Mallet, JB Filippi - Neural networks, 2021 - Elsevier

… Numerical **simulation** of **wildland fire spread** is useful to … of the burned surface returned after **simulation** of a fire igniting … After training on a large **simulation** dataset, the network shows …

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Developing the US **wildland fire** decision support system

EK Noonan-Wright, TS Opperman… - Journal of …, 2011 - Wiley Online Library

… , fire **behavior prediction**, economic assessment, smoke … The objective of this paper is to **review** the US **Wildland Fire** … environment in the context of **wildland fire management**. Next, we …

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AI for **Wildfire Management**: From **Prediction** to Detection, **Simulation**, and Impact Analysis—Bridging Lab Metrics and Real-World Validation

N Caron, HN Noura, L Nakache, C Guyeux, B Aynes - AI, 2025 - mdpi.com

… In this article, we **review** the main domains of **wildfire management** where AI has been … We first identify high-risk regions, **forecast** risk, detect predicted ignitions, simulate fire **spread**, and …

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